

1. snowflake schema consists of:

Fact Table: Hospital - Visit - Fact

Dimensions: Patient, Doctor, Department, Time (normalized into related hierarchy tables).

b) OLAP operations:

* Roll-up Time $\rightarrow$ Year

* Roll-up Patient $\rightarrow$ All

* Roll-up Doctor $\rightarrow$ All

* Keep department unchanged

Final Cuboid: [All, All, Department, Year]

c) Total cuboids: $4^4 = 256$

2. Smart City - Star schema

Star Schema

Fact Table: City - Fact

Measures:

* Traffic congestion level

* Vehicle count

* Energy consumption

* Punctuality %

Dimensions:

* Time (Year, Month, Day)

* Sensor

* Location (City, Zone, Road)

* Citizen

Justification: Measures store numerical data for analysis, while dimensions provide different views.

3. Banking Data Warehouse

a) Galaxy Schema

* Transaction - Fact (Transaction count, amount)

* Loan - Fact (Loan amount, interest)

Shared dimension:

* Customer

* Branch

* Time

* Account - Type

b) OLAP Operations

*) Roll-up Time $\rightarrow$ Year *) Roll-up Customer $\rightarrow$ All
*) Roll-up Account Type $\rightarrow$ All *) Keep branch

c) Total cuboids

Each dimension has 4 levels

$$4^4 = 256.$$

4. Manufacturing company

Roll-up: Product Category + Year + All Factories

$\rightarrow$ Defect Rate.

Drill-down: Factory $\rightarrow$ Department $\rightarrow$ Product type

$\rightarrow$ Defect Rate.

slice: Supplier Rating ≥ 4 $\rightarrow$ Production Units.

Dice: Selected product categories + Factory locations

$\rightarrow$ Defect Rate and Supplier reliability.

5. Retail chain

Dimensions:

*) Product (Category, Subcategory, SKU)

*) Store (Region, City, Store ID)

*) Time (Year, Quarter, Month, Day)

*) Customer (Membership Tier, Age group, gender)

Measures:

*) Inventory Turnover

*) Total Revenue (sales)

*) Units sold